

# ***ZEN High-Res Receiver™***

The new Zonge Electromagnetic Network (ZEN)™ receiver is a high-resolution, multi-channel receiver for acquisition of controlled- and natural-source geo-electric and EM data.



Designed for high-performance and reduced deployment costs in the collection of MT data, the new ZEN High-Res Receiver™ collects the highest resolution data currently possible and weighs less than 6 pounds (2.6kg).

Large-scale surveys suddenly become possible with smaller crews due to the ease of transport and placement of stations.

Complete support, field peripherals, software, and training are available.

## **Uses 32-bit analog-digital converter designed specifically for geophysical applications**

Signals not detected with conventional 16-bit or 24-bit receivers are easily detected with ZEN's 32-bit resolution, making surveys successful even in areas of cultural noise.

## **Distributed acquisition, wireless arrays**

With a truly distributed acquisition system, survey design and setup becomes easier. Each analog signal can be measured wherever topography and accessibility make it convenient, rather than where survey geometry dictates.

- Resistivity, Time/Frequency Domain IP, CR, CSAMT, Harmonic Analysis CSAMT (HACSAMT), AMT, MT; DC to 1024 Hz
- 1/64 to 1024 Hz Controlled Source / IP
- Use as an analog or borehole data logger

- Rugged, portable, and environmentally sealed
- Modular design for easy upgrades
- Fully compatible with Zonge transmitters, controllers, and other ancillary equipment
- Native GPS synchronization with transmitter
- ZigBee® Radio Control
- Broadband time-series recording
- 60 MHz ARM CPU
- 1 to 6 channels, user expandable
- One 32-bit ADC *per channel* for maximum speed and phase accuracy
- Auto gain setting and internal calibration
- Storage: 8 GB data per channel for data and program, sufficient to hold several days' data
- USB data transfer

# Specifications

## General

Broadband, multi-channel, multi-function digital receiver  
Frequency range: DC - 1024 Hz  
Number of channels: 1 to 6 (user expandable)  
Standard Survey capabilities: Resistivity, Frequency- and Time-Domain IP, Complex Resistivity, CSAMT (scalar, vector, tensor), Harmonic Analysis (CSAMT, Frequency-Domain EM, MMR, Magnetic IP, Magnetotellurics, Downhole Logging.  
Software language: C++  
Size: 27 x 24 x 13 cm (10.5 x 9.5 x 6")  
Weight: (without batteries and meter/connection panel): 6 channels: 3 Kg (6lb 9oz)  
Enclosure: Heavy-duty, environmentally sealed aluminum  
Power: 7-36V rechargeable batteries (external pack)  
ZEN power consumption: 0.7 Ah  
Temperature range: -40° to +50°C (-40° to +122°F)  
Humidity range: 5% to 100%  
Internal temperature sensors  
Time base: GPS Synchronization

## Displays & Controls

Power On-Off  
Color coded LEDs  
ZigBee® or USB Control from external computer

## Standard Analog

Input impedance: >10 M $\Omega$  at DC  
Board dynamic range: 180 db  
Minimum detectable signal: 20 nV  
Maximum input voltage:  $\pm 2.5$ V  
Automatic gain ranging in binary steps from 1 to 64  
Common-mode rejection at 1000 Hz: >100 db  
Phase accuracy:  $\pm 0.1$  milliradians (0.006 degree)  
Adjacent channel isolation at 100 Hz: >90 db  
Analog to digital converter (standard channel)  
Resolution: 32 bits  
Conversion time: 0.25 msec  
One A/D per channel for maximum speed and phase accuracy  
Analog connection via Pomona or 16-pin waterproof Mil-Spec connector

## Digital Section

Microprocessor: 60 MHz ARM processor per channel  
Mass Storage: 8 GB per channel. Data storage device with capacities to 16 GB/channel is optional.  
Serial ports: USB connection to each channel  
Distributed Control: Long-range mesh network (Unlicensed 2.4 GHz)

## Acquisition Software

MT, CR, RDIP graphical interfaces for Windows based computers  
External Control: Serial String based interface enables easy custom development  
Real-time programmable through download of BASIC scripts



Specifications subject to change without notice. 2015/01/23

Zonge International is an employee-owned company providing ground geophysical field services, consulting, and customized equipment to geoscientists and geotechnical engineers worldwide. The company is a pioneer in the development and application of broadband electrical and EM methods.

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